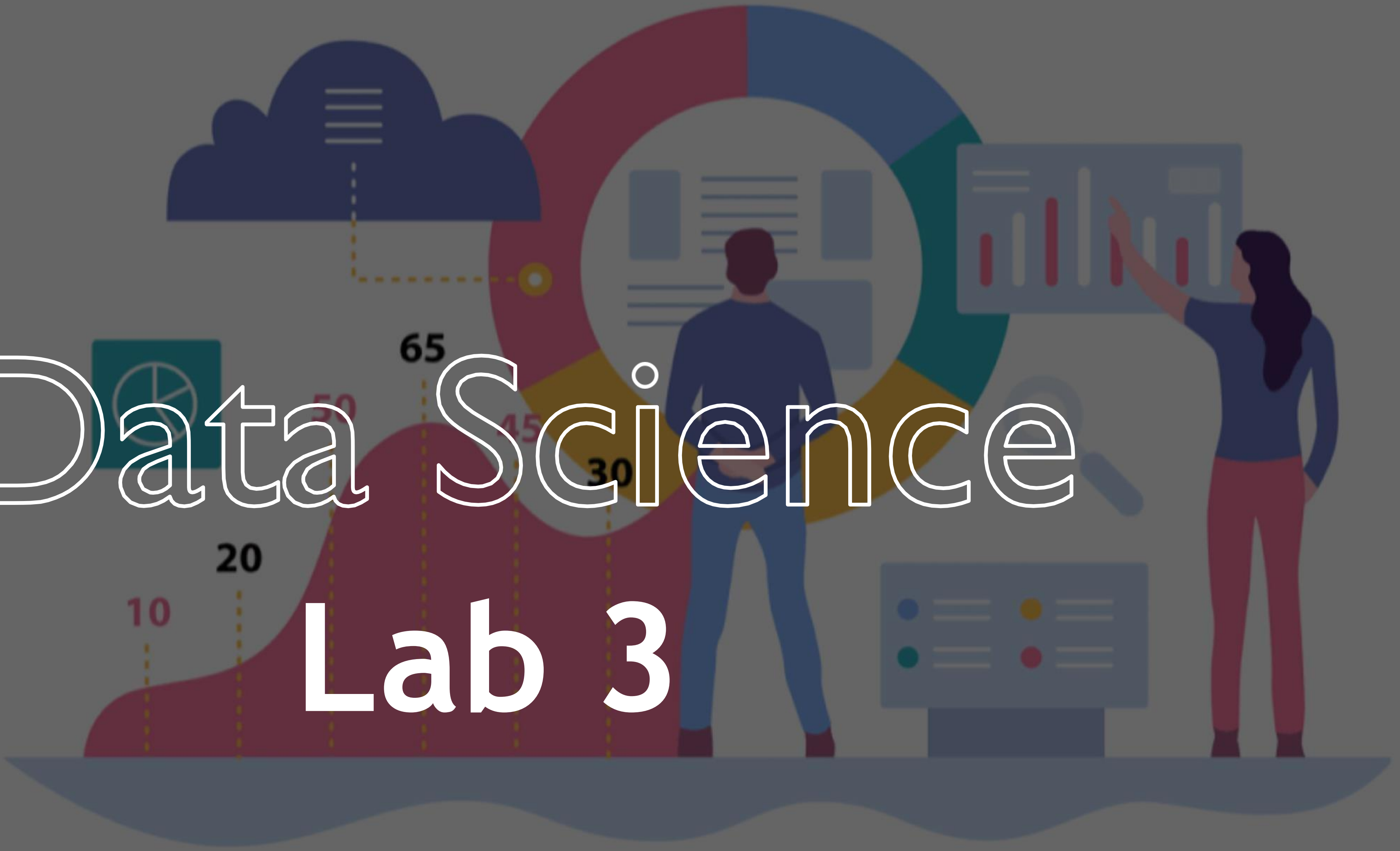
Data Science

Lab 3



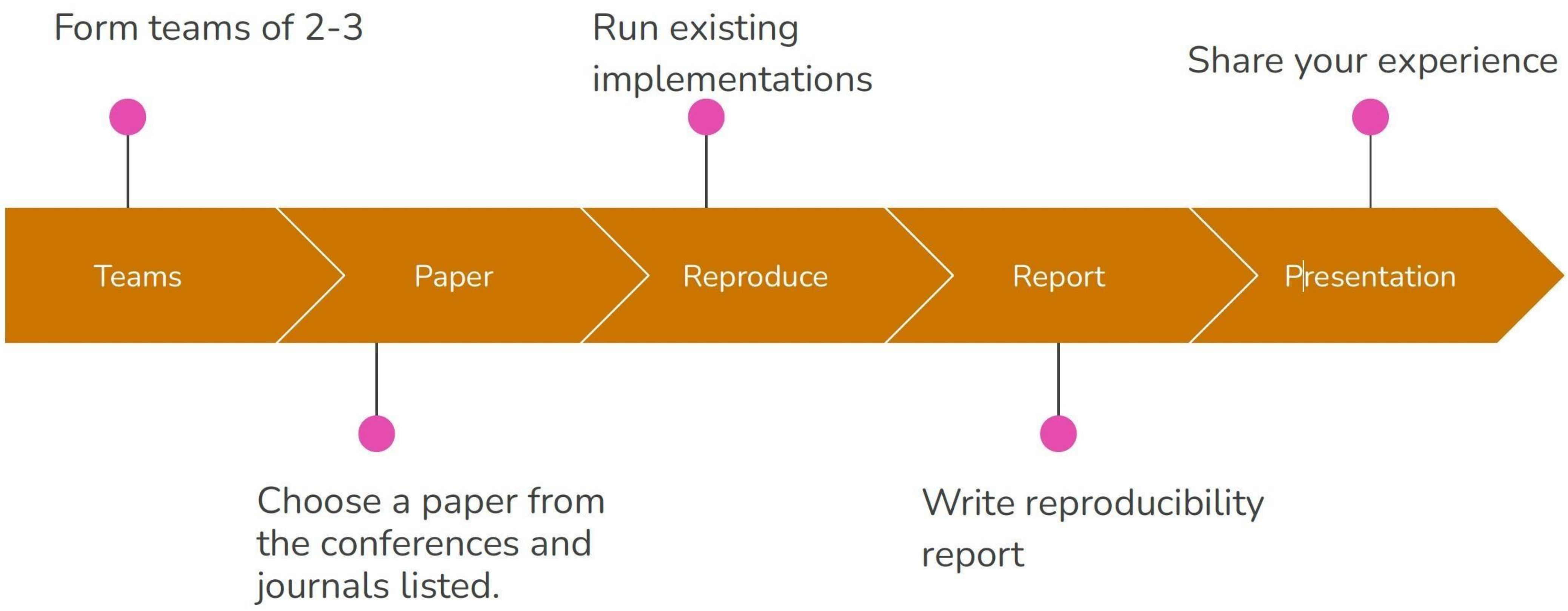
Objectives Reminder

- Understand ongoing research in the field of data science
- Identify research claims from papers
- Reproduce results using code/data provided by the authors

Reproduce Results Reminder

- Identify the main claims of the paper.
- Run the code provided by authors to validate the claims.
- Contact authors.
- Tune hyperparameters.

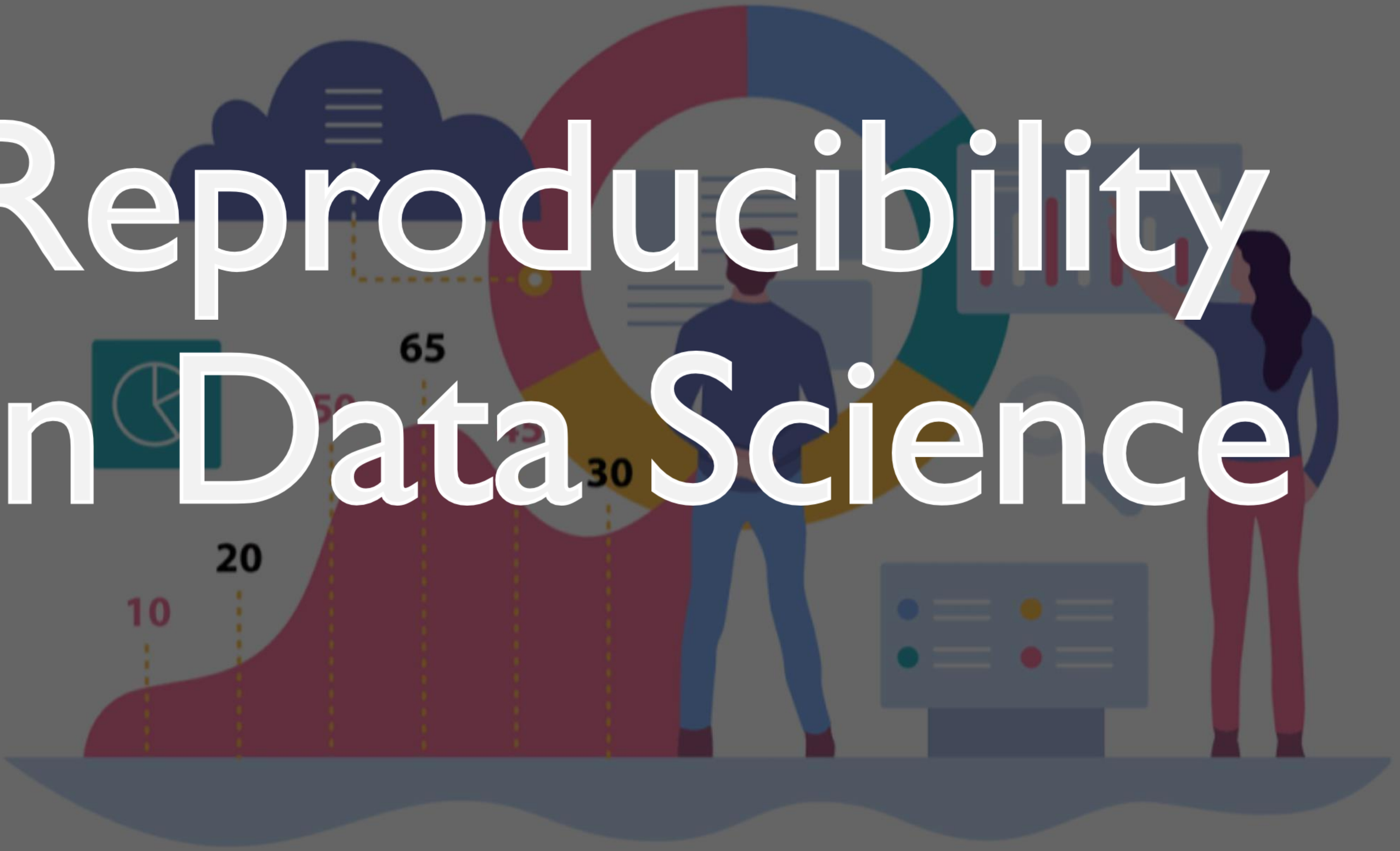
Lab 3 Task Reminder



Reproducibility Report & Presentation

- Template: Reproducibility Summary Template
- Maximum Length: 9 pages.
- GitHub Repository with the code.
- Discuss encountered challenges, along with recommendation.
- Deadline: **06.11.2024**.

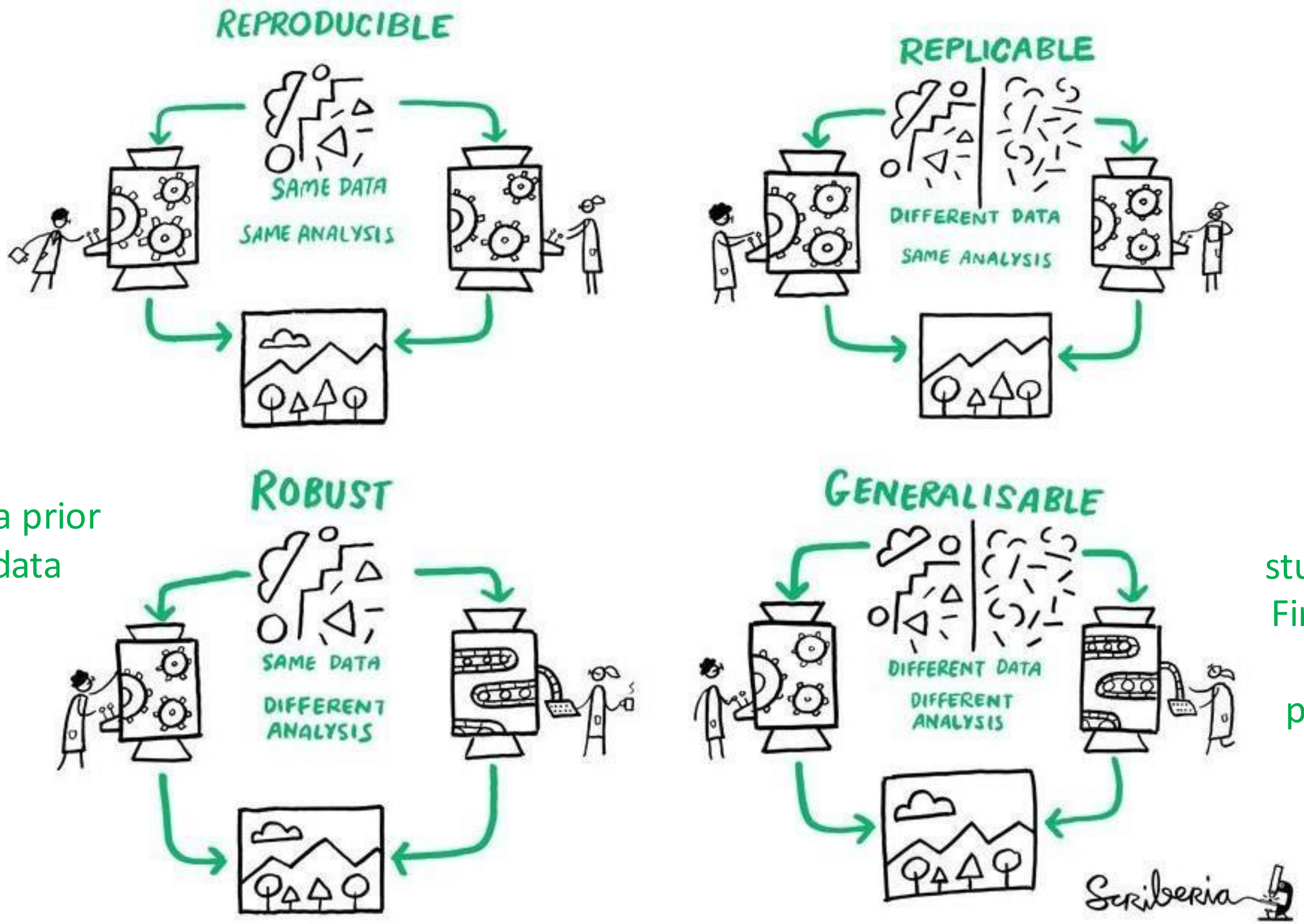
Reproducibility in Data Science



Reproducibility in Data Science

Achieving the same results using the same data and analysis as the original.

Yielding same results using the same methods but new data.



Testing the reliability of a prior finding using the same data and a different analysis strategy.

Applying the results of a study to a broader context. Findings can be applied to most contexts, most people, most of the time.

https://www.scribbr.com/

How to Begin Reproducing Paper Results

- How to choose the paper?
- How to manage the paper structure?
  - Abstract: short statement about the paper designed to give the reader a complete, yet concise understanding of the research questions and findings.
  - Introduction: a section which provides a broad overview of the approach often in a non-technical & reader-friendly manner. It is advisable to start by reading the introduction section before delving into the more intricate details of the research.
  - Related Work: in this section authors elaborate why their work is pertinent, how they address significant issues, and emphasize the originality. While this section holds immense value for the scientific community, practitioners can skip it.
  - Research Methods: explains the methods used to address the research query, describe in detail the algorithms employed, provide rationale for the chosen experimental design, data pre-processing techniques, detailed algorithmic design, training tricks, and post-processing.
  - Experiments: elaborates on the datasets used, training regimen, ablation studies, evaluation, hyperparameter tuning and comparison to state-of-the-art approaches.
  - Conclusion: summary, underlining the main contribution and future research directions.

Papers With Code

- Browse state-of-the-art:
  - Computer Vision, Natural Language Processing, Medical, ML Methodology, Time Series, Graphs, Speech, Reasoning, Playing Games, Computer, Code , Audio, Adversarial Methods, Robots, Knowledge Base, Music, etc.
- Explore Datasets:
  - Image,Text, Audio,Video, Medical, etc.
- Explore papers grouped by specific methods:
  - General, ComputerVision, NLP, Reinforcement Learning, Sequential Methods, etc.
- ML Reproducibility Challenge 2022

Phase I - Find a Paper

Papers with code - 20 min

Team discussion - 10 min

Phase II - Pitching

Prepare - 20 min

Pitch- group x 3 min